Introduction to Programming
COMP101

Tutorial-06
Pseudocode Iteration

Objectives:

Exercise-01 Pseudocode Iteration to Python

Exercise-02 Pseudocode to Iteration to Python

Introduction

Iteration (while loop, for loop) is a construct that repeats the flow of control of a program.
Most statements may execute once only (in sequence) or may not execute at all.

Pseudocode is intended to be not language-specific so even though our platform (Python) does use 'while loops' and 'for loops', we only use 'loop' and leave the actual implementation (for or while) to when we program it.
The general approach to identifying the logic of iteration across all programs that prevails.

Hence we refer to the language sub-set of our pseudocode to design the program, i.e. we use 'loop' and maybe 'loopend'

Exercise-01: Pseudocode Iteration

A die (singular of the plural 'dice') is a six-sided cube with each side of the cube showing an integer 1 to 6. There is one unique integer on each side.

A user 'throws' two die. If they throw a 'double six' they are told they can play the game again. This continues until a number other than 12 is thrown and play is passed to the next player.

#no initialisation

1. input
   - 1.1 input integer1
   - 1.2 input integer2

We are not trying to implement the program at this stage but to establish its logic ready for coding.

For instance, here we would probably code an import random function – but this is language specific and has no place here

2. process
   - 2.1 calc total = integer1 + integer2
   - 2.2 loop
     - 2.2.1 if total = 12 then
       - 2.2.2 print ‘Throw again’
       - 2.2.2 loop
         - 2.2.2.1 input integer1
         - 2.2.2.2 input integer2
       - 2.2.3 loopend
     - 2.3 else
       - 2.3.1 print ‘pass dice’
   - 2.5 loopend

Nested loop

3. Output
   - 3.1 print game ended

Notes Exercise-01:

a) No initialisation included due to trivial design with limited i/o.
Depending on the logic and the specification, it may not need to include all sections of the design

b) We don't state what type of loop this is – it could be a for loop or a while loop but these are implementation issues.
In another language it could be a repeat-until loop (which is not in Python)

c) Each 'loop' has a 'loopend' – not compulsory but in the example it makes it clearer to see where the inner loop ends and where the outer loop ends – which may help clarify how to implement it